Cloud Computing

Digital Assignment - 01

21MIS1132
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1) How can VIT University implement different types of cluster computing systems, such as HPC clusters, load balancing clusters and storage clusters to enhance the efficiency & reliability of its academic, research and administrative operations?

⇒ **Implementation of Cluster computing systems at VIT university:**

Cluster computing refers to the use of multiple interconnected computers that work together as a single system to enhance computational power, efficiency and reliability. VIT university can implement different types of cluster computing systems to support various academic, research and administrative functions.

Types of Cluster Computing Systems:

1) **High-Performance computing Clusters**

→ **Purpose & Benefits:**

HPC clusters are designed to perform large-scale computations and simulations, making them essential for research intensive universities. These clusters can support:

* **Scientific Research:**

Simulations in physics, chemistry

and engineering that require high-speed computing.

* Machine learning and AI:
Training deep learning models on large datasets.

* Bioinformatics & Genomics:
Processing genetic sequences and medical research data.

* Materials Science:
Simulating molecular structures and nanotechnology applications.

→ Implementation at VIT:

* Hardware & Software
  - High-speed interconnected nodes with CPUs and multi core GPUs.
  - Storage solutions like RAID configurations, SAN/NAS systems. Linux-based operating systems.
  - High-speed networking via InfiniBand or 40/100 Gbps Ethernet. Parallel processing libraries like MPI and OpenMP for distributed computing.

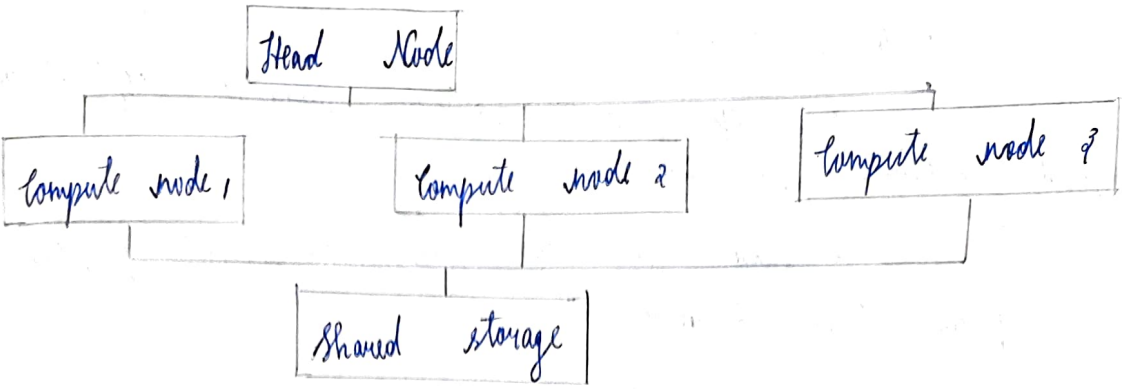
→ Expected outcomes:

* Faster and more efficient execution of computation tasks.

* Increased research output and collaboration with global institutions.

* Reduced time for simulations & modeling.

→ HPC Cluster Architecture:



ii) Load-Balancing Clusters:

→ Purpose and benefits:

Load-balancing clusters distribute workloads across multiple servers to enhance the reliability and responsiveness of web-based applications. This is crucial for:

* Learning Management System (LMS):

Platforms like moodle, which handle thousands of student logins simultaneously.

* Online portals:

Managing admission processes, result publication and fee payments.

* Digital libraries and Research Databases:

Providing smooth access to students and faculty.

→ Implementation at VIT :

* Infrastructure :

• Multiple web servers running applications and database servers.

• Load balancers such as HAProxy, Nginx, or Apache Load Balancer.

• Cloud - based CDN services for content delivery.

* Load balancing techniques :

• Round Robin - requests are distributed evenly across available servers.

• Least connections - directs requests to the server with the fewest active connections.

• IP Hashing - directs users to the same server based on their IP for consistency.

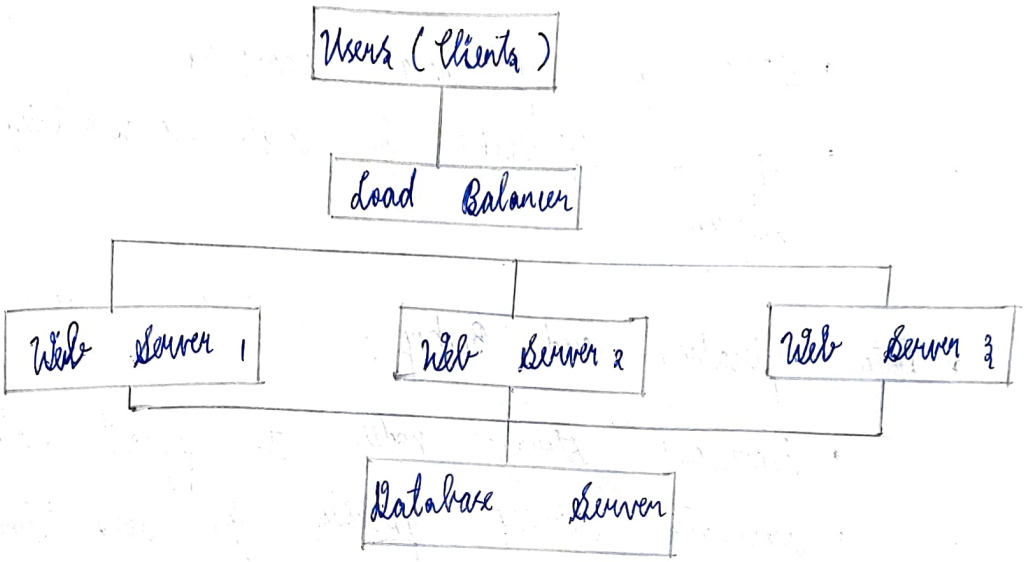
→ Expected Outcomes :

* Prevents system crashes during peak usage periods.

* Ensures uninterrupted access to critical services

* Enhances the overall user experience.

→ Load - Balancing Cluster Architecture :



iii) Storage Clusters :

→ Purpose and Benefits :

Storage clusters provide scalable, fault - tolerant and high - availability storage solutions, enabling VIT to manage vast amounts of data effectively. This includes :

* Student records :
Managing enrollment data, grades and attendance.

* Research data :
Storing large datasets from ongoing research projects.

* E - learning content :
Hosting lecture videos, assignments and reference materials.

→ Implementation at VIT:

* Distributed File Systems:
  - HDFS for big data analytics
  - Ceph or GlusterFS for high-availability storage.

* Data replication and Backup:
  - Redundant storage nodes to prevent data loss.
  - Automated backup policies for long-term data retention.

* Cloud-based storage solutions:
  Integration with platforms like AWS S3, Google Cloud Storage for additional scalability.

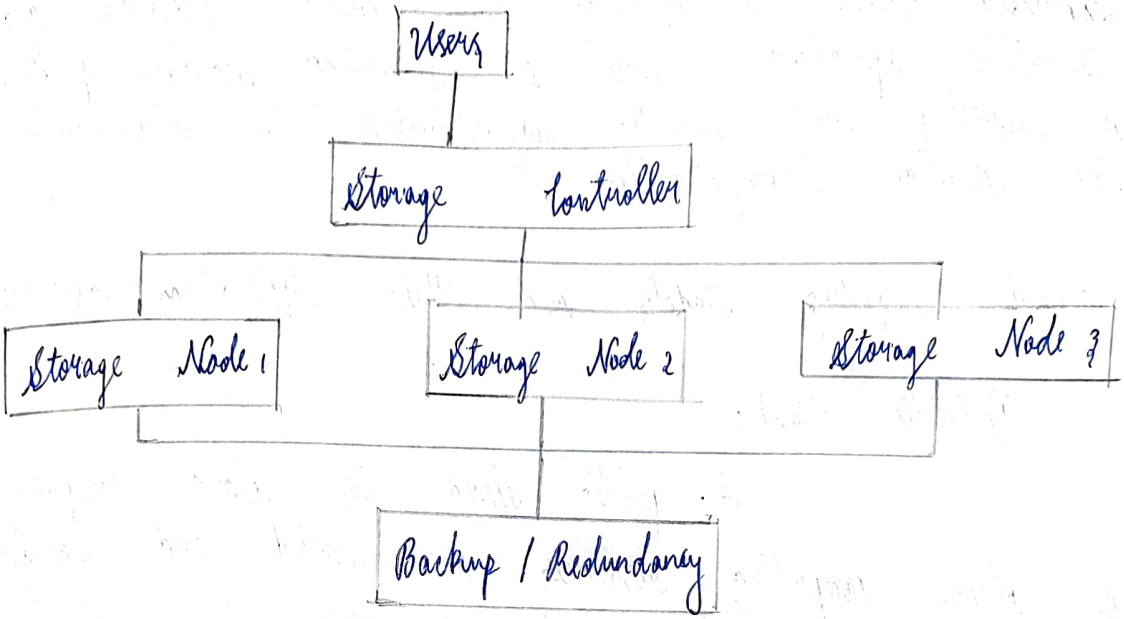
→ Expected Outcomes:

* Secure and reliable storage for critical data.

* Easy accessibility and quick retrieval of academic materials.

* Scalability to accomodate growing university data needs.

→ Storage Cluster Architecture :



Conclusion :

Thus by implementing HPC, load-balancing and storage clusters, VIT University can significantly improve its computational efficiency, reliability and scalability. These solutions will not only enhance research capabilities but also ensure seamless academic and administrative operations.

2.) How can VIT University leverage different types of cloud computing models - public, private, hybrid and community clouds - to optimize its academic, research and administrative operations, such as providing scalable platforms and ensuring cost-effective infrastructure for university-wide digital transformation?

→ Cloud computing models and their applications at VIT

i) Public Cloud:

A public cloud is cloud computing model where computing resources are hosted and managed by third-party providers such as AWS, Google Cloud and Microsoft Azure.

→ Uses cases at VIT:

* Scalable E-learning platforms:

Hosting learning management systems such as Moodle on cloud-based virtual machines and using cloud storage to store lecture recordings and assignments.

* Research and AI/ML model training:

Running AI and Machine learning models on cloud platforms like Google Cloud AI or AWS SageMaker and using HPC instances to process large datasets.

* Administrative and student services:

Hosting high-traffic student admission portals to handle thousands of applications in real time and using secure cloud-based payment gateways to process student fees.

→ Advantages:

* Scalability - easily handles peak loads
* Cost Efficiency - pay per use model
* Reliability - ensures 99.99% uptime
* Global Accessibility - access resources from anywhere

→ Challenges:

* Data Security Risks
* Internet Dependency

ii) Private Cloud:

It is a cloud infrastructure dedicated solely to VIT University. The university can host data and applications on its own servers or use managed private cloud services like OpenStack, VMware or IBM Private.

→ Use cases at VIT:

* Secure Student and Administrative Data Management:

Storing student records, financial transactions and confidential research data on private servers and implementing role-based access to restrict access to sensitive files. Ensuring compliance with data protection regulations.

* Examination and Academic records:

Managing question papers, exam results and degree certificates securely and implementing blockchain-based credential verification to prevent fraud.

* Faculty and Staff portals:

Hosting HR management systems, payroll processing and faculty evaluation platforms.

→ Advantages:

* Enhanced security - firewall, encryption & VPN access
* Customization - tailored infrastructure
* Regulatory compliance - meets data privacy laws

→ Challenges:

* Higher costs - Investments in hardware & maintenance
* Limited Scalability

iii) Hybrid cloud:

A hybrid cloud combines the strengths of both public and private clouds, allowing VIT to maintain sensitive data on-premise while leveraging the scalability and cost-effectiveness of the public cloud.

→ Use cases at VIT:

* Storing critical data privately while using public cloud for scalability:

Storing student information, confidential research data and financial records on the private cloud and using public cloud resources to run simulations, host online exams or provide computing power for AI/ML projects.

* High - performance computing for research:

Training AI models and running simulations on the public cloud and storing highly sensitive research data locally to ensure security.

* Cloud Bursting for handling high traffic:

During peak seasons, the university website and portals can dynamically scale using public cloud resources while maintaining primary services on a secure private cloud.

→ Advantages:

* Optimized costs
* Security and Scalability
* Flexible Resource Allocation

→ Challenges:

* Complex Management
* Data Synchronization Issues

iv) Community Cloud:

It is a shared cloud infrastructure where multiple universities and research institutions collaborate on joint projects. This model enables resource sharing, cost efficiency and knowledge exchange.

→ Use cases at VIT:

* Shared Research Infrastructure:

Partnering with universities and research labs to share AI / ML models, datasets and computing resources. Hosting joint research projects in various domain sectors.

* Consortium - based digital libraries:

Accessing and sharing research papers, journals and historical archives through cloud - based digital repositories.

* Secure Collaborative Learning Environments:

Virtual classrooms where students and faculty across institutions can collaborate. Hosting inter - university hackathons and innovation challenges.

→ Advantages:

* Cost reduction
* Encourages Academic Collaboration
* Centralized Resource Management

→ Challenges:

* Governance and Ownership
* Security Concerns